Agentic SOC: A Hierarchical Swarm-Orchestrated Multi-Agent LLM Architecture for Autonomous OT Cyber Defense

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***Abstract*— The convergence of enterprise IT networks with operational technology (OT), industrial control systems (ICS), and 5G/6G wireless-enabled cyber-physical infrastructures has expanded the attack surface of critical environments, exposing limitations in traditional IT-centric Security Operations Centers (SOCs). IT–OT systems impose strict latency, safety, and availability constraints, while rule-based detection and human-driven triage lead to alert fatigue, delayed response, and fragmented visibility across wired and wireless domains. This paper presents Agentic SOC, a hierarchical swarm-orchestrated multi-agent LLM architecture for autonomous and adaptive cyber defense in IT–OT converged networks. The architecture decomposes SOC operations into three cooperative tiers: a perception layer of distributed agents performing real-time telemetry analysis across IT, OT, and wireless network domains; a reasoning layer executing cross-domain correlation, MITRE ATT&CK (including ICS) mapping, and coordinated mitigation; and a strategic leader agent conducting kill-chain reconstruction, RAG-grounded threat hunting, and adversarial simulation. We implement and evaluate Agentic SOC under a managed security service provider (MSSP) model across four safety-critical scenarios: 6G telecom infrastructure, a power generation plant, a railway control system, and a three-dimensional Security of Air–Ground–Space Integrated Networks (SAGIN) environment. Experimental benchmarking demonstrates broader threat coverage, improved cross-domain correlation, reduced false positives, and a lower mean time to detection and response compared to flat, multi-agent, and rule-based SOC baselines while preserving human-in-the-loop governance. Agentic SOC provides an experimentally validated blueprint for AI-native, self-protecting SOC architectures in next-generation IT–OT cyber-physical systems.**

Keywords— Hierarchical Multi-Agent Systems, Swarm Intelligence, Autonomous Cyber Defense, IT-OT Security, 5G/6G Wireless Security, Security Operations Center

I. Introduction

The rapid digital transformation of critical infrastructure has accelerated the convergence of enterprise Information Technology (IT) networks with Operational Technology (OT), Industrial Control Systems (ICS), and emerging 5G/6G-enabled cyber-physical systems [1]. Historically isolated, these domains now interconnect through shared networks, remote access mechanisms, edge computing, and software-defined control platforms [2]. While this convergence improves efficiency and visibility, it also expands the attack surface. Adversaries increasingly exploit enterprise IT footholds or wireless access points to pivot into OT environments, targeting programmable logic controllers (PLCs), industrial protocols, and safety-critical control loops through kill chains that traverse wireless, IT, and OT boundaries. Unlike IT systems, OT environments prioritize availability and operational continuity, as disruption can lead to physical damage or safety hazards.

Security Operations Centers (SOCs) are responsible for monitoring, detecting, and responding to threats across organizational networks. However, conventional SOC architectures are largely IT-centric, relying on centralized SIEM systems, static correlation rules, and manual analyst triage. Many OT systems rely on legacy infrastructure with limited patch cycles, minimal encryption in industrial protocols, and incomplete segmentation. Strict latency and determinism requirements constrain intrusive inspection, and centralized SOC pipelines become bottlenecks that generate excessive false positives and struggle to correlate multi-stage attacks spanning wireless access networks, IT infrastructure, and multiple Purdue levels.

The Purdue Enterprise Reference Architecture (PERA) [3] structures industrial control environments into hierarchical levels ranging from physical processes (Level 0) to enterprise systems (Level 5). Although widely adopted for network segmentation, traditional SOC monitoring does not fully align with this layered structure, with limited distributed intelligence across control and supervisory layers. As IT–OT convergence reduces separation between levels, there is a need to extend the Purdue Model with coordinated, distributed defensive intelligence.

Recent advances in Large Language Models (LLMs) have demonstrated utility in log interpretation, alert summarization, contextual reasoning [4], and intelligent network management [5]. However, deploying a single centralized LLM within a SOC pipeline introduces challenges: high-volume heterogeneous telemetry can overwhelm centralized reasoning, latency-sensitive OT operations cannot tolerate unpredictable decision delays, and automated actions affecting physical systems require structured governance.

These limitations reveal a research gap: existing SOC automation approaches do not adequately support distributed intelligence across Purdue levels, structured coordination among multiple reasoning components, or continuous integration of defensive and adversarial validation in wireless-enabled IT–OT converged systems. To address these challenges, this paper introduces Agentic SOC, a hierarchical swarm-orchestrated multi-agent LLM architecture for autonomous cyber defense in IT–OT infrastructures that enhances the Purdue Model by embedding coordinated AI agents across different architectural levels.

The architecture decomposes SOC functionality into three cooperative tiers. At Tier 1, swarms of lightweight agents operate near telemetry sources across IT and OT domains, performing distributed anomaly detection and first-pass threat screening. At Tier 2, supervisory agents validate candidate

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findings, perform cross-domain correlation, map behaviors to MITRE ATT&CK (including the ICS matrix), and orchestrate mitigation actions. At Tier 3, a strategic leader agent aggregates global intelligence, reconstructs multi-stage attack campaigns, conducts RAG-supported threat hunting, and executes adversarial simulations. Swarm coordination is enabled through structured agent-to-agent (A2A) communication protocols and Model Context Protocol (MCP)-based tooling, allowing bounded-context reasoning at each tier while ensuring explicit handoff, escalation, and human authorization workflows.

We implement and evaluate Agentic SOC across four heterogeneous cyber-physical scenarios: a 5G/6G telecom infrastructure [6], a power generation plant, a railway signaling system, and a three-dimensional SAGIN environment [7]. Comparative evaluation against rule-based SOC architectures and flat LLM-agent baselines demonstrates improved cross-domain correlation, broader MITRE ATT&CK coverage [8], reduced false positives, and faster validation cycles.

The main contributions of this work are as follows:

1. *Hierarchical Swarm-Orchestrated SOC Architecture for IT–OT Convergence.*

   We propose Agentic SOC, a hierarchical multi-agent LLM architecture that extends the Purdue Model by embedding distributed defensive intelligence across IT–OT layers, enabling scalable and layered cyber defense beyond centralized SOC designs.

2. *Structured Agent Coordination via A2A and MCP for Governed Autonomy.*

   We introduce a structured agent-to-agent (A2A) communication model combined with Model Context Protocol (MCP)-based workflows to support bounded-context reasoning, explicit escalation, and human-in-the-loop authorization for safety-critical infrastructures.

3. *Causal TTP Attribution and Campaign-Level Predictive Reasoning.*

   We develop mechanisms that map telemetry to MITRE ATT&CK (including ICS) techniques and infer adversarial progression from partial kill chains, enabling proactive defense rather than purely reactive response.

4. *Agentic AI-Assisted Red Teaming Integrated into the SOC Control Fabric.*

   We implement an agentic red teaming platform capable of generating MITRE-aligned TTPs and executable attack scripts for controlled VAPT. Unlike traditional testing, our approach embeds adversarial validation directly within the SOC architecture, enabling continuous assessment of detection rules and analytics pipelines.

5. *Closed-Loop Blue–Red Defense Validation Across MSSP Environments.*

   We demonstrate a unified blue–red architecture evaluated across four heterogeneous managed services such as 5G/6G telecom, power plant, railway control, and SAGIN systems showing improved threat coverage, reduced false positives, and accelerated validation cycles compared to rule-based and flat AI baselines.

6. *Confidence-Calibrated Human-in-the-Loop Governance.*

   We introduce structured confidence scoring and evidential grounding mechanisms that ensure AI-generated defensive actions remain transparent, accountable, and safely governed in mission-critical IT–OT deployments.

The rest of the paper is organized as follows. Section II presents the proposed multi-tier Agentic SOC and the AI-driven red team framework architecture and implementation. IT, OT, and cyber-physical environments. Section III discusses the evaluation results. Finally, Section IV concludes the paper and outlines future research directions.

II. System Architecture and Implementation

Agentic SOC implements a hierarchical multi-agent framework organized into multiple tiers that collectively realize autonomous offensive and defensive security operations.

A. Defensive System

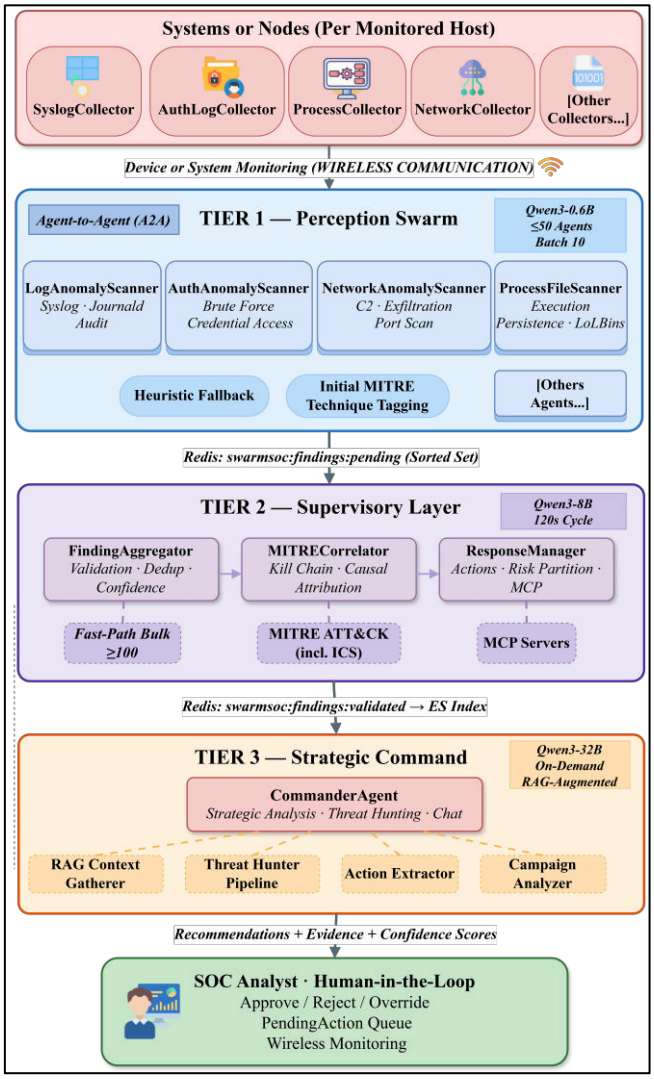


Fig. 1. Three-tier Agentic SOC defensive architecture showing endpoint data collection, Tier-1 perception swarm for anomaly detection, Tier-2 supervisory correlation and response orchestration, and Tier-3 strategic command with human-in-the-loop and wireless-communication layers.

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Tier-1 Perception Swarm: Tier-1 instantiates lightweight perception agents operating near data sources including wireless edge nodes such as 5G MEC servers and IoT gateways, performing distributed first-pass anomaly detection on top of rule-based filtering. These agents employ small language models (0.5B–1B parameters) optimized for rapid inference. The system dynamically spawns scanner swarm instances based on different types of telemetry requirements. Specialized scanner agents include the LogAnomalyScanner for syslog and journal audit analysis, AuthAnomalyScanner for brute force and credential access detection, NetworkAnomalyScanner for C2 exfiltration [9] and port scan identification, ProcessFileScanner for execution and persistence detection via living-off-the-land binaries, and WirelessProtocolScanner for detecting rogue access points and abnormal radio-layer signaling [10]. Each scanner produces initial MITRE ATT&CK technique tagging with heuristic fallback mechanisms. Findings are pushed to a Redis sorted set (swarmsoc:findings:pending) for consumption by the supervisory layer. The generalization of these agents allows integration in various types of IT-OT scenarios.

Tier-2 Supervisory Layer: Tier-2 deploys supervisory agents with 8-14B parameter models providing deeper reasoning for validation, correlation, and response orchestration, operating on a 120-second cycle. The layer comprises three specialized agents.

The FindingAggregator agent consumes pending findings from Tier-1 using fast-path bulk processing (batches of 100 or more); applies confidence scoring separating true positives from false alarms, adjusts severity based on environmental context, and enriches findings with validation notes. Findings confirmed as legitimate threats promote to validated status, while false positives filter out reducing analyst burden.

The MITRECorrelator agent performs cross-domain event fusion with AI-powered causal attribution. Unlike traditional SIEM correlation that identifies only temporal or spatial coincidence, Agentic SOC employs causal attribution, inferring likely relationships between attack techniques and observable telemetry patterns. The system maintains a knowledge base mapping MITRE ATT&CK techniques (including ICS) to expected telemetry signatures. LLM reasoning evaluates whether observed events match expected causal chains considering temporal ordering, process ancestry, privilege contexts, wireless signaling patterns, and network relationships, transforming isolated anomalies into coherent cross-domain threat narratives.

The correlator also implements intent-aware campaign reasoning, maintaining campaign state tracking and detected techniques mapped to ATT&CK tactics. When new findings emerge, LLM reasoning evaluates current campaign progression against known attack patterns to predict the adversary's next moves. This predictive capability enables proactive defensive positioning, for instance, preemptively increasing monitoring on high-value targets or deploying deception assets along predicted attack paths.

The ResponseManager agent generates defensive action recommendations partitioned by risk level. Low-risk operations execute automatically under policy constraints, while high-impact actions, including host isolation, process termination, and firewall modifications, queue for explicit human approval with trust-calibrated explainability. All recommendations include structured explanations with evidential grounding and confidence calibration.

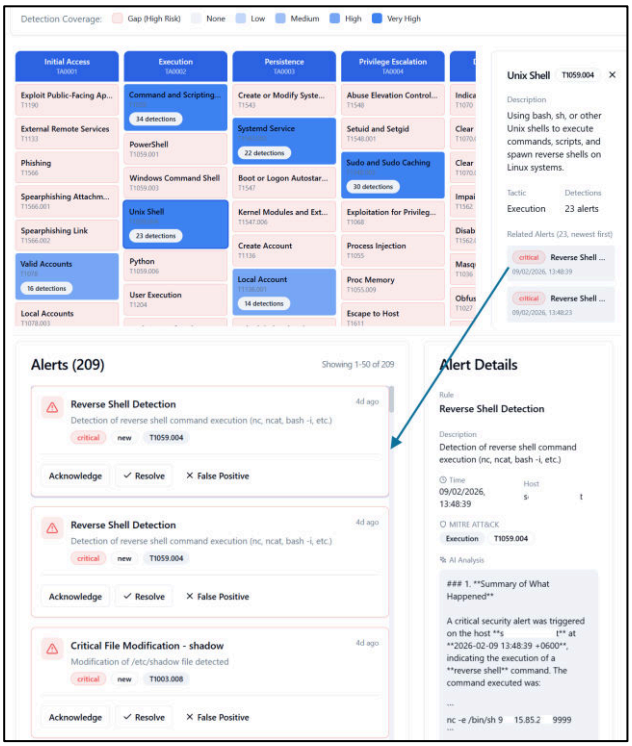


Fig. 2. MITRE ATT&CK–aligned alert visualization showing detection coverage across tactics and an example of reverse shell alert (T1059.004 – Unix Shell) with AI-assisted analysis and analyst response options.

Tier-3 Strategic Command: Tier-3 instantiates the Commander AI using a large open-weight agentic model (30-50B parameters), providing sophisticated reasoning for on-demand threat hunting and defensive operation planning. The commander agent implements evidence-centered retrieval augmented generation, aggregating recent events, alerts, validated findings, and statistical summaries across wireless and wired domains to answer analyst queries with grounded recommendations. Responses include structured evidence objects referencing specific telemetry, confidence scores calibrated to available information, and actionable next steps appropriate to threat context.

The threat hunting pipeline decomposes proactive investigation into specialized stages: the planner formulates hunt hypotheses based on environment-specific threat intelligence and environmental risk; the evidence retriever queries telemetry with adaptive refinement; the anomaly detector identifies deviations from expected patterns; the correlation analyst establishes relationships between anomalous behaviors and known attacks; the hypothesis ranker prioritizes findings by threat severity and confidence; the response strategist recommends containment; and the hunt reporter generates structured documentation. This swarm-like multi-agentic staged pipeline enables systematic threat discovery beyond reactive alert processing. Fig. 3 illustrates the multi-agentic threat hunting interface.

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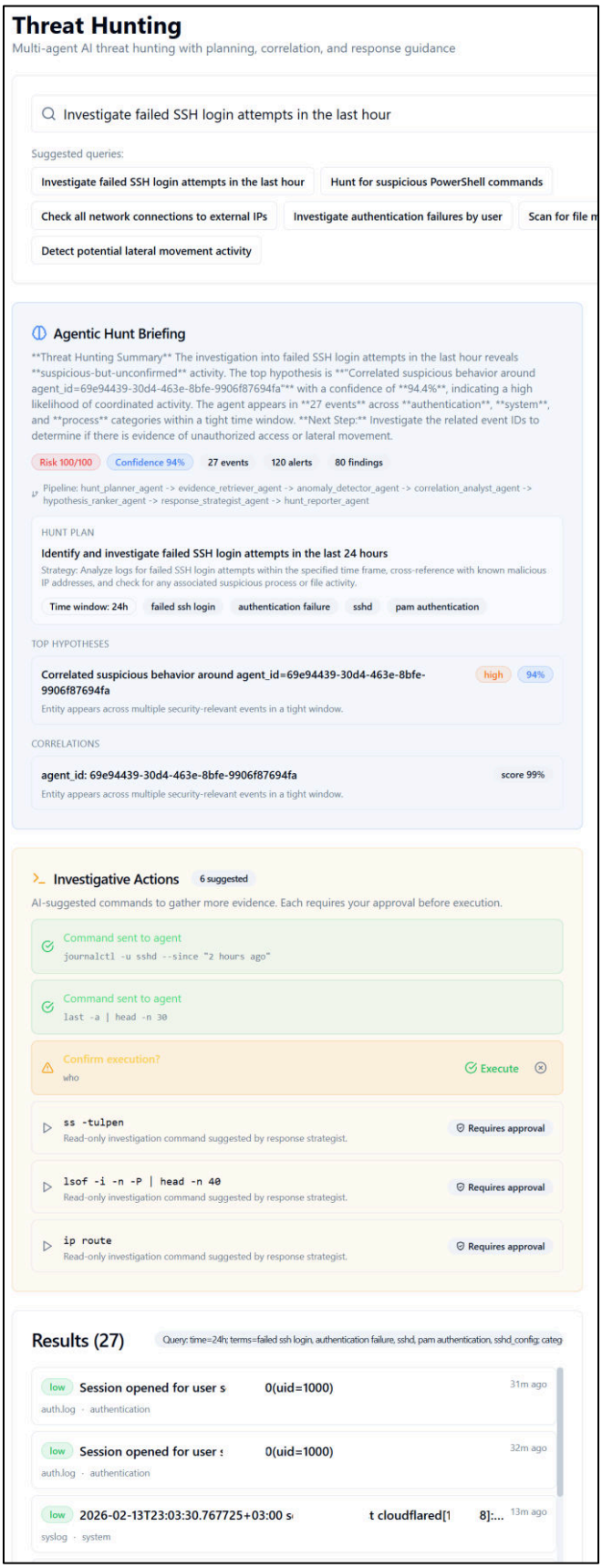


Fig. 3. Multi-agentic threat hunting interface showing AI-generated hunt briefing, correlated hypotheses, risk and confidence scoring, and human-approved investigative command execution scenario for failed SSH login analysis.

B. Offensive System

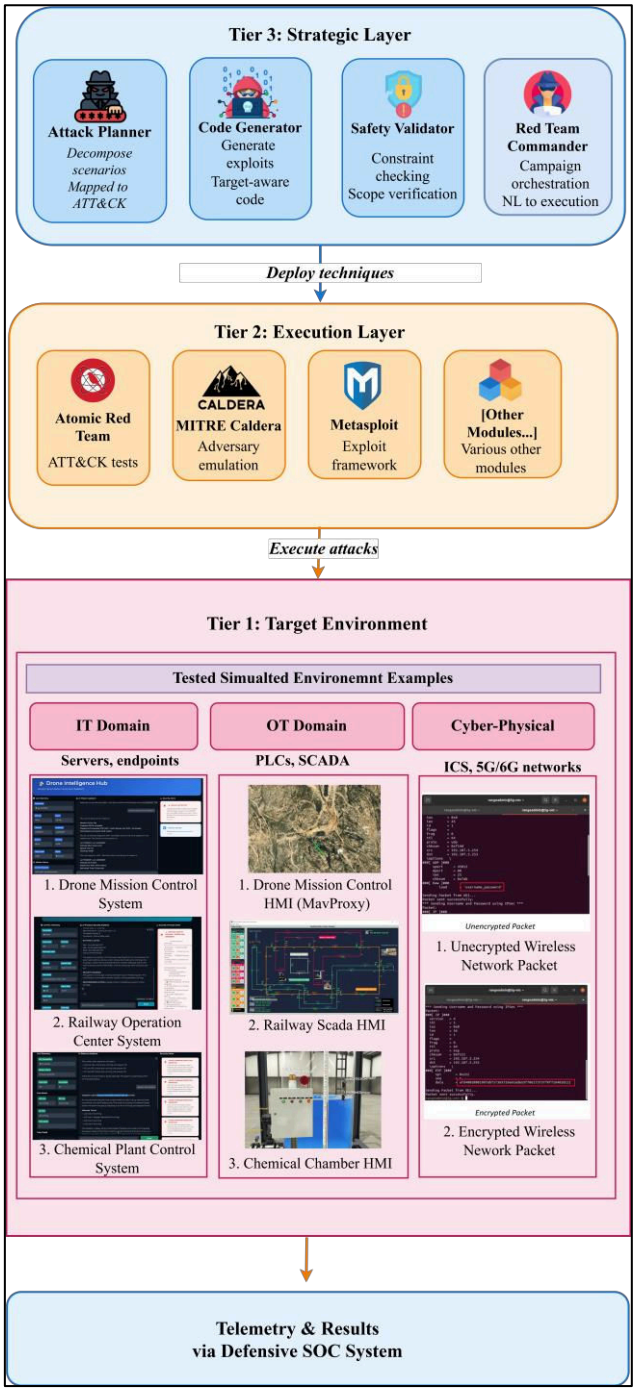


Fig. 4. Three-tier AI-driven red team architecture showing strategic planning and safety validation (Tier 3), automated adversary emulation and exploit execution tools (Tier 2), and deployment across IT, OT, and cyber-physical test environments with telemetry fed back to the defensive SOC system.

The offensive operation adopts a three-tier hierarchical structure analogous to the defensive architecture; however, in this configuration, the agents are prompt-engineered to execute red-teaming activities rather than defensive functions. It comprises a modular and extensible toolkit including platforms such as Atomic Red Team, Metasploit, MITRE Caldera, and custom adversarial simulation modules each enhanced with AI-driven capabilities to enable strategic, automated offensive operations and adaptive attack orchestration. Fig. 4 illustrates the three-tier offensive architecture.

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The strategic layer (Tier 3) implements semantic-to-operational threat compilation, translating natural language attack hypotheses into executable offensive pipelines. When analysts specify techniques to test, the system handles planning, validates corpus availability and target readiness, and then executes tests while collecting telemetry.

The execution layer (Tier 2) integrates established adversary emulation frameworks, including Atomic Red Team for validated ATT&CK technique tests, MITRE Caldera for adversary emulation campaigns, and Metasploit for exploit execution. These modules deploy techniques against target environments spanning IT domains (servers, endpoints), OT domains (PLCs, SCADA), and cyber-physical networks (ICS, 5G/6G networks).

For novel scenarios lacking available modules, the strategic layer employs generalized LLM-based offensive command and code generation. Analysts describe scenarios in natural language, which are decomposed into ATT&CK-mapped technique sequences, generated into target-aware executable code, validated for safety constraints, and executed with telemetry collection. This capability democratizes adversarial testing, enabling security teams to rapidly validate defenses against arbitrary threat scenarios without requiring deep offensive expertise. Fig. 5 illustrates the red team interface demonstrating automated generation and execution of HTTP-based data exfiltration commands with interactive MCP action control.

Telemetry and results from all offensive operations feed back into the defensive SOC system, creating a closed-loop blue–red validation cycle that enables continuous assessment of detection coverage and response effectiveness.

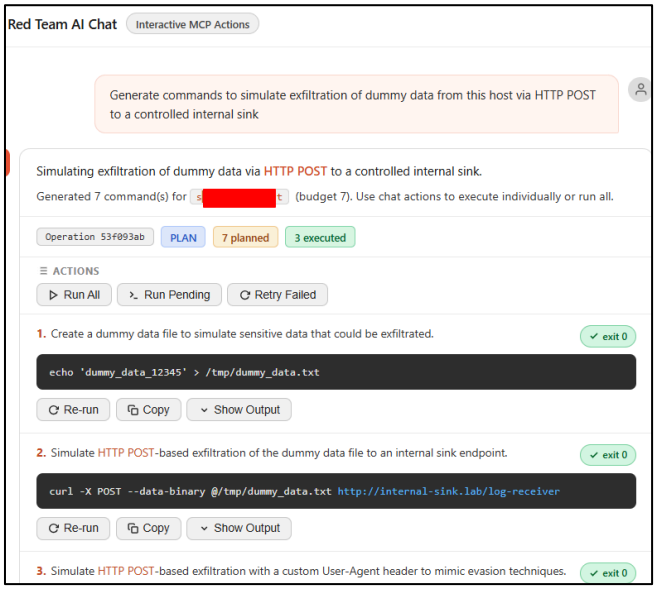


Fig. 5. AI-assisted red team operation interface demonstrating automated generation and execution of HTTP-based data exfiltration commands with interactive MCP action control and stepwise execution tracking.

C. Human-in-the-Loop Governance

Autonomous defensive actions require careful governance balancing automation efficiency with operational safety. Agentic SOC implements risk-partitioned human-in-the-loop workflows via Model Context Protocol (MCP) tooling providing structured interaction between AI agents and human analysts. Low-risk actions, including read-only investigations and data collection, execute automatically under policy constraints. High-impact actions including host isolation, process termination, and network modifications generate approval requests presented to analysts with full context: threat assessment, recommended action, expected impact, and rollback procedures. Analysts approve, modify, or reject recommendations, with decisions recorded for audit and learning. This governance model keeps human control over important decisions while automating everyday tasks, striking a balance between quickness and safety that is crucial for launching production.

III. Experimental Setup And Evaluation

A. Evaluation Methodology

We evaluate Agentic SOC using 107 Atomic Red Team techniques spanning all twelve MITRE ATT&CK tactic categories, representing realistic attack patterns in Linux-based IT-OT environments. An automated pipeline executes each test in a controlled environment, captures telemetry (process execution, network connections, file modifications, and system commands), and feeds it into both detection configurations for comparative analysis.

B. Comparative Configuration Design

Configuration A: Rule-Only Detection. This baseline relies exclusively on signature-based detection rules using pattern matching logic with field-level checks, regular expressions, and logical operators, representing traditional SIEM approaches.

Configuration B: Rule+AI Detection. This configuration augments rule-based detection with the three-tier hierarchical agent architecture. Telemetry first passes through the rule engine, then undergoes AI-powered analysis via the Tier-1 perception swarm, Tier-2 supervisory correlation, and selective Tier-3 strategic reasoning. The AI layer examines both rule-matched events and events that failed rule matching, applying contextual analysis, behavioral profiling, and causal reasoning.

C. Detection Performance Results

Table I presents the comprehensive detection performance comparison across all evaluated techniques, organized by MITRE ATT&CK tactic.

TABLE 1. Detection Performance By MITRE ATT&CK Tactic

Tactic	Total	Rule	Rule%	AI	AI%	Δ
Defense Evasion	32	14/32	43.8%	27/32	84.4%	+13
Discovery	23	4/23	17.4%	18/23	78.3%	+14
Privilege Escalation	10	10/10	100.0%	10/10	100.0%	0
Credential Access	10	5/10	50.0%	8/10	80.0%	+3
Command and Control	8	6/8	75.0%	7/8	87.5%	+1
Collection	7	6/7	85.7%	6/7	85.7%	0
Impact	6	4/6	66.7%	5/6	83.3%	+1
Exfiltration	4	2/4	50.0%	3/4	75.0%	+1
Execution	3	2/3	66.7%	3/3	100.0%	+1
Persistence	2	1/2	50.0%	2/2	100.0%	+1
Initial Access	1	1/1	100.0%	1/1	100.0%	0
OS Credential Dumping	1	0/1	0.0%	0/1	0.0%	0
TOTAL	**107**	**55/107**	**51.4%**	**90/107**	**84.1%**	**+35**

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Configuration A (Rule-Only) achieved 52.3% detection (56/107), while Configuration B (Rule+AI) achieved 84.1% (90/107), a +31.8 percentage point improvement. Figure 6 presents a visual comparison of detection performance.

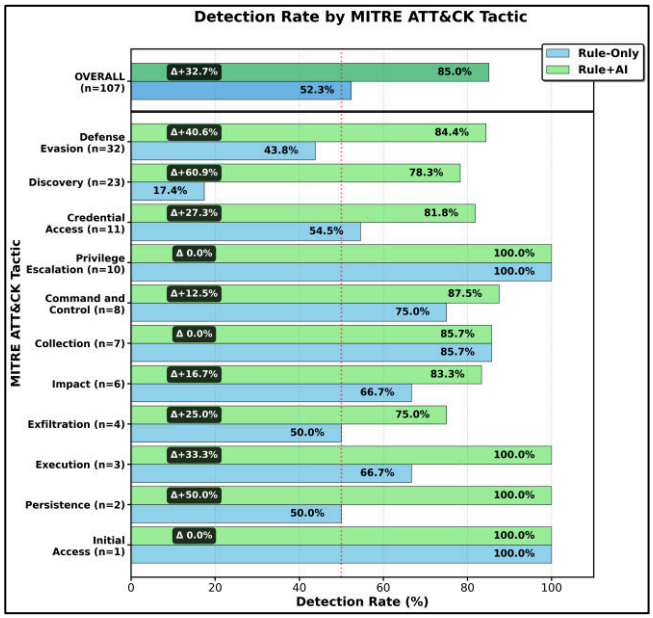


Fig. 6. Detection rate comparison between Rule-Only and Rule+AI approaches across MITRE ATT&CK tactics. Rule+AI achieves 85.0% overall detection versus 52.3% for Rule-Only, demonstrating a 32.7 percentage point improvement (n=107).

Discovery techniques increased from 17.4% (4/23) to 100% (23/23) detection, an +82.6-percentage point improvement. These techniques generate telemetry resembling benign administrative activity, making signature-based detection challenging. Defense Evasion improved from 43.8% (14/32) to 90.6% (29/32), a +46.9% gain. The 34 techniques newly detected by AI concentrate heavily in Discovery (19) and Defense Evasion (15), confirming that hierarchical agent intelligence provides maximum benefit for contextual, behavior-based detection.

D. Discussion

The evaluation demonstrates that hierarchical multi-agent LLM architectures can substantially improve detection coverage when integrated into existing SOC workflows. The +31.8% improvement validates the core hypothesis: distributed perception agents (Tier 1) combined with supervisory correlation (Tier 2) and strategic reasoning (Tier 3) can detect contextual attack patterns that evade signature-based rules. Most importantly, improvements concentrate in Discovery and Defense Evasion, tactics most frequently exploited by APTs, where traditional detection faces fundamental challenges.

IV. Conclusion and Future Works

This paper presented Agentic SOC, a hierarchical swarm-orchestrated multi-agent LLM architecture for autonomous cyber defense in IT–OT converged networks. The architecture decomposes SOC operations into three cooperative tiers i.e., perception, supervisory, and strategic command enabling scalable and layered cyber defense that extends beyond centralized SOC designs. Structured agent-to-agent (A2A) communication and Model Context Protocol (MCP)-based workflows ensures bounded-context reasoning with explicit escalation and human-in-the-loop authorization for safety-critical environments.

Experimental evaluation using 107 Atomic Red Team techniques (Linux) across twelve MITRE ATT&CK tactic categories demonstrated that Agentic SOC's AI-augmented filtering achieves a weighted average detection rate of 84.1% compared to 52.3% for rule-based filtering alone, representing a 31.8% improvement. The integrated offensive system with agentic red teaming enables closed-loop validation of detection coverage, moving beyond traditional periodic penetration testing.

Future work will extend Agentic SOC evaluation across heterogeneous cyber-physical scenarios, including 6G telecom infrastructure, power generation plants, railway control systems, and SAGIN environments under a managed security service provider (MSSP) model. We plan to incorporate reinforcement learning for adaptive agent policy optimization, explore federated learning for privacy-preserving cross-organization threat intelligence sharing, and develop formal verification methods for safety-critical autonomous response actions. We also intend to evaluate the system against advanced persistent threat (APT) campaigns with extended dwell times and multi-stage lateral movement patterns to assess campaign-level reasoning effectiveness in operational deployments. Additionally, we will benchmark Agentic SOC against other AI-driven SOC to provide reproducibility baselines and enable researchers to assess cost-effective deployment options for industrial environments.

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